RealQM for H_2 and He

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Abstract

RealQM replaces the many-electron wavefunction by N one-electron charge densities on non-overlapping domains, interacting through Coulomb's law with each pair counted once. At one electron it is the minimal model that Coulomb's law admits—a kinetic penalty quadratic in $\nabla\Psi$ against the kernel attraction, nothing else. Its *form* is forced, but the coefficient of the kinetic term is a scale the model cannot derive, and hydrogen's observed spectrum is what fixes it. Hydrogen therefore *calibrates*—the energy scale of the whole construction, and separately the one constant the discrete scheme needs—and cannot test anything.

H_2 is that same model extended to two electrons with *no further calibration*, in the two forms the interface condition admits: imposed strongly on a sharp partition, or weakly on a diffuse one. Both are predictions in the strict sense—nothing has been fitted at the two-electron level—and both agree fairly with observation, each on one observable: the strong form places the bond within 3.7%, the weak form the binding energy within 7.7%, while each is substantially worse on the other, and the two straddle the observed binding energy from either side. Helium is reported for the strong form only: the weak form returns a state that fails the sphericity test, for an implementation reason we identify, and we decline to quote an energy from it. The model acquires content at two electrons, where the densities must be partitioned and must meet, and we examine the two smallest such systems: H_2, whose interface lies between the kernels, and He, whose interface passes through one. Two defensible interface treatments—strict homogeneous Neumann on a sharp partition, and amplitude continuity enforced weakly on a diffuse one—*bracket* the observed energies of both systems from opposite sides. We argue that this interval, and not either endpoint, is what the construction predicts, and that the width of the interval is the honest statement of its accuracy at two electrons.

1 The model

RealQM models the ground state of an atomic system—kernels of charge Z_a at positions $\mathbf{R}_a$, surrounded by N electrons—in terms of a wave function

$$\Psi \;=\; \sum_{i=1}^{N} \psi_i \;\in\; H^1(\mathbb{R}^3), \qquad \int_{\Omega_i} \Psi^2 = 1 \;\text{ for each } i, \tag{1}$$

in which $\psi_i = \Psi|_{\Omega_i}$ is the restriction of Ψ to a domain Ω_i, the Ω_i being disjoint and filling $\mathbb{R}^3$, as the minimiser of the energy

$$E[\Psi, \{\Omega_i\}] \;=\; \tfrac{1}{2}\int |\nabla\Psi|^2 \;-\; \int K\,\Psi^2 \;+\; \sum_{i<j} \iint \frac{\psi_i^2(\mathbf{x})\,\psi_j^2(\mathbf{y})}{|\mathbf{x}-\mathbf{y}|} \;+\; \sum_{a<b} \frac{Z_a Z_b}{R_{ab}}, \tag{2}$$

with $K = \sum_a Z_a/|\mathbf{x}-\mathbf{R}_a|$ the kernel attraction, the minimisation being over *both* Ψ and the partition $\{\Omega_i\}$—the domains are unknowns of the problem, not given data. The only inputs are the kernel charges, their positions, and N. The domains being disjoint, $\Psi^2 = \sum_i \psi_i^2$ with

no cross terms, so the third sum is a genuine pair interaction counted once, and no electron interacts with itself.

Three features of this statement are worth drawing out, because the rest of the paper turns on them.

Continuity is not imposed; it is the function space. Ψ is required to lie in H^1, so it is continuous across every interface by assumption, and no matching condition is added by hand. What *is* free is where the interfaces are.

The interface condition is a consequence, not an input. Varying Ψ at a *fixed* partition leaves a boundary term $\oint \psi_i \, \partial\psi_i/\partial n$ on each $\partial\Omega_i$, so the natural condition is $\partial\Psi/\partial n = 0$ from each side—nothing is imposed. Solved that way the two one-sided values at an interface generically disagree, and the interface is then placed where they agree, which is what varying the *partition* supplies. Three conditions, one extra unknown; the count closes. The two implementations in §3 are two discretisations of this one variational problem and differ only in how they represent Ψ near the interfaces—not in the problem being solved.

There is one function, not N of them, and it is a scalar function in three dimensions. The decomposition (1) is bookkeeping: ψ_i is a restriction of Ψ, not an independent orbital, and the Ω_i are the only thing distinguishing one electron from another. So the unknown is a single scalar field on $\mathbb{R}^3$ whatever N may be, and at a spatial resolution h it is represented by a mesh of h^{-3} points on a unit cube—some 10^3 values per a_0^3 at the $h = 0.1\,a_0$ used here, independent of the number of electrons.

That is the whole of why this model is computable, and it is worth stating as the arithmetic rather than as a claim. The cost of a calculation grows with N only through the number of domains to be tracked and the pair sum in (2), not through the dimension of the unknown, so adding an electron adds work linearly rather than multiplying the size of the space to be searched. Every number in this paper was produced in a web browser, which would not be possible otherwise.

Two mechanisms, doing two different jobs. It is worth separating them, because they are easily run together. The Coulomb term does *not* keep two electrons apart: for smooth densities $\iint \rho(\mathbf{x})\rho(\mathbf{y})/|\mathbf{x}-\mathbf{y}|$ is finite even when the two coincide, so it permits coincidence at a finite price and cannot forbid it. What forbids it is the *partition*, which is a constraint and not a penalty: the domains are disjoint by construction, so overlap is unavailable rather than expensive. What the Coulomb term does is resist *compression*—as two domains are pushed together V_{ee} rises, and that sets where the interface sits and how density distributes inside each domain.

The distinction matters for what follows. Between *equivalent* domains across a mirror plane the partition contributes nothing beyond what Coulomb already supplies: the sharp and continuous constructions of §3 agree there to 4×10^{-8}. So in exactly that configuration the construction's whole resistance to compression is the classical Coulomb repulsion of two charge clouds, and the over-binding reported in §5 is the measurement that this alone is not enough. H_2 at its midplane and He at its split plane are both that configuration, which is why they are the systems to examine.

What is new, and what is missing. Relative to one electron the construction adds exactly two things: the densities do not overlap, and they must meet somewhere. It also *subtracts* one: it fixes what resists their approach, and the answer is set out below—the partition makes overlap impossible while the Coulomb term alone resists compression. That is a property of

the model, stated in its own terms and checkable within it, and it is why no two-electron system should be expected to come out exactly.

2 One calibration, on hydrogen

At one electron (2) is $\frac{\kappa}{2}\int|\nabla\Psi|^2 - \int(Z/r)\Psi^2$ minimised subject to $\int\Psi^2 = 1$. A kinetic penalty quadratic in $\nabla\Psi$ is the lowest-order rotation-invariant term that prevents collapse and no simpler functional binds at all, so the *form* is forced; but the coefficient κ is not. It is a scale, and the model contains nothing from which to derive it.

The coefficient of the kinetic term is the model's one physical calibration, and hydrogen sets it. Requiring the minimiser to reproduce hydrogen's observed spectrum fixes $\kappa = 1$ in atomic units, i.e. the familiar $\frac{1}{2}\int|\nabla\Psi|^2$, whereupon the minimiser is $\Psi = \sqrt{Z^3/\pi}\,e^{-Zr}$ with energy $-Z^2/2$. We are explicit that this is a calibration and not a derivation. The construction contains *one* kinetic coefficient and nothing from which to compute it: no mass appears anywhere in (2), and no constant of action, so there is no product of such quantities to be supplied from elsewhere and no separation of κ into factors that the model could make. Hydrogen's spectrum fixes the single number, and in doing so sets the energy scale of everything that follows. That is the sense in which the one-electron model is *calibrated* rather than predictive, and it is why no agreement at one electron can be evidence for the construction.

With κ fixed, the continuum model is complete and contains no further adjustable quantity. Its discrete realisation needs one more constant, of a different kind: a grid cannot represent $1/r$, so the kernel carries a softening, $K = Z/\sqrt{r^2+\varepsilon^2}$ with $\varepsilon = \sigma h$. We fix σ once, by requiring the one-electron answer to be exact. In the axisymmetric solver used below this gives

$$\sigma = 0.2846 \;:\qquad E(\mathrm{H}) = -0.500227 \quad (0.045\%), \tag{3}$$

and in the three-dimensional solver $\sigma = 0.627$, read off a four-point ladder (-0.508456, -0.500962, -0.493833, -0.486591 at $\sigma = 0.4, 0.6, 0.8, 1.0$, linear in σ to better than a percent) and then verified at the production grid, where it returns $E(\mathrm{H}) = -0.499901$, an error of 0.02%. The two values differ because the two codes discretise differently; each is fixed by its own one-electron exactness, which is the point.

That same one-electron row carries a null test worth stating, because it costs nothing. A single domain has no interface, so any interface measure evaluated on it must vanish; the quantity $\int u^2|\nabla w|\,dV$, which is the regularised form of $\oint_\Gamma u^2 dS$ used in §5, returns 8×10^{-4} here against 0.417 for H_2. A measure that did not vanish on one electron would be measuring something other than an interface.

The mesh is declared, and σ is a function of it. Two things must be separated here. The spacing $h = 0.1\,a_0$ is *declared*: it is fitted to nothing, chosen once, and used for every system in this paper. The softening is then *determined* rather than chosen—at each h, the condition $E(\mathrm{H}) = -1/2$ fixes it, and at $h = 0.1$ that gives $\sigma = 0.627$. So the discrete scheme carries one declared number and one determined one, and strictly the determined one is $\sigma(h)$: a different mesh requires re-solving the same condition, not reusing 0.627. As h decreases the grid resolves the kernel better and less softening is needed, so $\varepsilon = \sigma(h)\,h \to 0$—the scheme does not freeze a finite softening into the continuum limit.

What this does *not* claim is that $h = 0.1$ is converged. It is not, and the paper says so where it matters: the weak construction's bond error falls $34.9 \to 28.3 \to 21.4\%$ at $h = 0.1395, 0.10, 0.08$ and is still falling. Every number below is therefore reported at a declared mesh and not at a limit, which is the honest status of a model whose results depend on h and whose h-dependence is measured rather than assumed away.

Where $h = 0.1$ comes from, and what precision to expect. The spacing is not chosen on any physical ground. It is the finest at which these systems compute in a *common web browser*: at box $20\,a_0$ that is 200^3 cells, and a two-electron run at that size already takes some tens of minutes on ordinary hardware. The whole simulator is deliberately a browser program—anyone can load it and reproduce every number here without installing anything—and $h = 0.1$ is the resolution that constraint permits.

The consequence should be stated plainly rather than discovered by the reader: **the precision to expect is fair, not high.** Several of the errors reported below are mesh-limited rather than model-limited, and we say which where we can tell them apart—helium's energy most clearly, where the softening moves the answer by nine percent and the quantities of physical interest are smaller than that. Where an h-ladder exists we give it instead of a single figure, and where the ladder has not flattened we say so. A finer mesh is the obvious next step and is not available under the constraint we have adopted.

This is the second constant, and the last. Three properties of it matter. It is performed on a *one-electron* system, where the model has no content of its own and the reference is exact, so no two-electron observable is consumed. It is a single number, not a function of anything. And it is fixed before any two-electron system is computed. Everything below follows with no further calibration.

What this is not. It is not an optimisation of ε against a residual. That operation is available and must be refused: the total force residual is monotone in ε and minimised by unphysical settings—hydrogen ceases to bind at $\varepsilon = 0.75h$—so a scheme that tunes ε to improve an error measure drifts away from the physics it is meant to describe. Fixing it on a known energy is a different operation and has a unique answer.

3 The interface, and the two constructions

For a fixed partition, free variation of each density in (2) leaves a boundary term $\oint u_m\, \partial u_m/\partial n$, so the natural condition is homogeneous Neumann, $\partial u/\partial n = 0$, on each domain's own boundary—nothing is imposed. Solved that way the amplitudes at the interface generically *disagree*, and the interface is then placed where they agree. Continuity is therefore not a second assumption about Ψ; it is the equation that determines Γ. Three conditions, one extra unknown.

Two implementations each supply one half of that scheme and take the other half from elsewhere.

(A) Sharp partition, strict Neumann. Labelled cells, with the kinetic flux through any face joining cells of different ownership *omitted at assembly*. Omitting a face *is* the homogeneous-Neumann condition—zero flux implemented by not adding a flux, with no extra equation. This imposes the condition exactly but requires Γ to be known, which for H_2 and He is supplied by symmetry. One trap is worth recording: taking a *submatrix* of the full stiffness instead retains the diagonal contributions of faces leading into the excluded region and so imposes Dirichlet, pushing density off the boundary and under-binding. Measured on a one-electron problem outside a core, 3.977 against 6.177 eV; the latter matches a one-dimensional radial solve to 2%.

(B) Diffuse partition, continuity enforced weakly. An occupancy w_m evolves on the local amplitude mismatch $c = u_m - \max_{n\neq m} u_n$ and converges where that mismatch vanishes, so this construction genuinely *finds* Γ—the only one of the two that does. Its Neumann condition is only natural, not imposed: varying $\frac{1}{2}\int w|\nabla u|^2$ makes $w\,\partial u/\partial n = 0$ stationary, but on a finite grid a face straddling the interface carries half flux rather than none. That half-flux is

an extra kinetic cost sitting exactly at the interface, and it is a first-order discretisation error, not a structural one: H_2's bond error falls $34.9 \to 28.3 \to 21.4\%$ at $h = 0.1395, 0.10, 0.08$ and is still falling.

The two are not a choice between theories. They are two incomplete implementations of one: (A) imposes the condition and is told where Γ is; (B) finds Γ and imposes the condition weakly. Combining them—binary faces on a *diffuse* occupancy—does not work and should not be attempted: it gives H_2 a binding energy of $+110\%$ and a dipole of 0.108–0.187 D where reflection symmetry requires identically zero, because closed faces make an amplitude jump at the interface free. The diagnostic is direct: the kinetic energy of the assembled field $\Psi = \sum_m w_m u_m$, which charges that jump, exceeds the per-domain sum by a factor of 8.07, so the discrete solution is not in H^1.

4 H_2: two kernels, interface between them

Hydrogen fixed one constant and settled nothing else. H_2 is the same model carried to two electrons *without any further calibration*: the same σ, the same functional, the same $i \neq j$ Coulomb rule, and the measured separation as the only other input. Every number in this section is therefore a prediction in the strict sense, and that is the only reason the agreement below is worth reporting—a two-electron constant fitted here would have made it worthless.

It is also the mildest two-electron system available. Each kernel is $Z = 1$, so the softening calibrated on hydrogen applies unchanged to both and nothing is transferred from a different kernel charge; and the interface sits at the midplane by reflection symmetry, in the *tail* of both densities rather than near either nucleus. The extension admits exactly two forms, differing only in how the interface condition is imposed—strongly, on a sharp partition, or weakly, on a diffuse one—and we report both.

With σ fixed by (3) and the atomic reference measured at the *same* σ—a requirement, not a nicety, see below—the binding energy comes out

$$E_{\text{bind}} = 2E(\text{H}) - E(\text{H}_2) = 2(-0.499901) - (-1.160855) = 0.161053,$$

against $D_e = 0.174470$: **under-binding by** 7.7%, at $R = 1.5\,a_0$ on the commensurate grid nearest the observed 1.40028. The sharp construction at its own calibration gives $+59.6\%$ at its own minimum while placing the bond within -3.7%. The two therefore fall on *opposite sides* of the observed binding energy.

The bond, which is the better-behaved observable. The separation at which each construction's energy is lowest is a prediction carrying no reference at all beyond the measured $R_e = 1.40028$. The sharp construction puts it at $1.349\,a_0$, -3.7%, from a parabola through three commensurate points about the minimum. The weak construction at $\sigma = 1$ gives $1.8 \pm 0.1\,a_0$ ($+28\%$), the uncertainty being the width of the band within which the energy differences fall below the $\pm 0.0018\,E_h$ initialisation scatter. At the calibrated $\sigma = 0.627$ its minimum has not yet been located: a sharper kernel deepens the well and should shorten the bond, so we expect it between those two figures, but we have one point and not a scan, and report the gap rather than interpolate it.

The ordering is worth noting because it is the opposite of the energy's. On the bond the sharp construction is the accurate one, within four percent; on the binding energy it is the inaccurate one, at $+59.6\%$ against the weak split's -7.7%. No single construction here is better on both.

Two implementations, complementary errors, and what that does and does not license. The two differ in *exactly one* respect—how the interface condition is imposed—and everything else about them is the same model, the same calibration and the same geometry. So the complementarity is not a coincidence of two unrelated methods: the weak construction's error is a half-flux localised *at the interface*, which for a diatomic is precisely where the bond length is determined, and it is therefore large in the bond and much smaller in the binding energy. That the pair gets one observable right each is a statement about where the deficiency lies, and it localises it to the interface treatment alone.

We are careful about what this licenses. Quoting the sharp bond together with the weak binding as "the model's answer" would require a rule, settled in advance, for which construction to trust for which observable. For the bond there is such a rule: H_2's interface position is fixed by reflection symmetry, so the sharp construction's *only* handicap—that Γ must be supplied to it from outside—is inactive here, and its geometry is accordingly the one to quote. For the binding energy no corresponding argument is available: the weak construction's handicap is never inactive, so its -7.7% is not privileged over the sharp $+59.6\%$ by anything beyond being nearer the measurement, and selecting it on that ground would be fitting. The pair therefore *diagnoses*; it does not predict both observables at once, and we do not present it as doing so.

A warning earned the hard way. The reference must be computed at the same σ as the molecule. Paired instead with an atom measured at $\sigma = 1$, the same H_2 run reports $E_{\text{bind}} = 0.188011$, i.e. $+7.8\%$ and apparently *over*-binding—the sign inverted by a mismatched reference. Since σ sets the cusp resolution and hence every energy, a reference at a different σ is not a reference at all.

5 He: hard split and weak split

Helium is the same two-electron question asked under far worse conditions. The kernel is $Z = 2$, so the softening calibrated on hydrogen is being extrapolated—though not unreasonably: the one-electron ion He^+ at the same σ gives -2.053249 against the exact -2.0, only 2.66% deep, so the cusp itself is well handled. And the interface passes *through* the kernel rather than lying between two of them, so it cuts the density maximum: $\int_\Gamma u^2 dS = 1.681$ against H_2's 0.496, a factor of 3.4.

Helium is special: both electrons touch the kernel. In every molecule below, each electron has its own kernel and the interface lies *between* them, in the tail of both densities. Helium has one kernel and two electrons, so a dividing surface necessarily passes *through* it and both domains touch it. The interface therefore cuts the density maximum rather than the tail: $\int_\Gamma u^2\, dS = 1.681$ against H_2's 0.496, a factor of 3.4. Helium is not a smaller molecule; it is a different configuration, and its two admissible constructions are not two approximations to one answer.

Two constructions, describing different things. The *homogeneous* treatment puts both electrons in one domain sharing one shape. For a two-electron closed shell that is exactly the restricted Hartree–Fock ansatz—one spatial shape, one Coulomb pair, which the $i \neq j$ rule supplies directly—so it should return the Hartree–Fock energy, and the shortfall against the exact value is the correlation energy that neither carries. The *split* treatment gives each electron its own territory, with Γ the plane through the kernel and the condition imposed exactly:

$$E(\text{He}) = -3.026749 \quad \text{against} \quad -2.903724, \qquad \textbf{over-binding by } 4.2\%.$$

What the homogeneous treatment measures. At the softening calibrated on hydrogen it returns $E = -2.771193$, with the state clean on every available check—$\mu = 0.000\,\mathrm{D}$ as sphericity requires, the norm exact at 2.000, and the interface measure 3×10^{-4} since there is no interface. The term breakdown is the informative part. Against the Hartree–Fock values $T = 2.8617$, $V_{eK} = -6.7492$, $J = 1.0258$:

σ	T	V_{eK}	V_{ee}
0.627 (calibrated on H)	2.5318 (−11.5%)	−6.3228 (−6.3%)	**1.0197** (**−0.6**%)
0.2846	3.1780 (+11.0%)	−7.3139 (+8.4%)	1.1135 (+8.6%)

At the calibrated σ the *pair* term is right to 0.6% and the whole error lies in the one-electron terms—which is where He^+ already placed it, with a 5.7% kinetic deficit at the same σ. So the two-electron content is sound and the unresolved $Z = 2$ cusp at $h = 0.1$ is what the energy is reporting.

And at this mesh the softening dominates helium. Moving σ from 0.627 to 0.2846 moves E by $0.251\,E_h$, about 9%, from 3.2% shallow of Hartree–Fock to 5.6% deep of it. Against that, the correlation energy is 1.45% of the total and the difference between the two constructions is smaller still. Three consequences follow and we state them rather than work around them. Helium's energy at $h = 0.1$ is a statement about resolution, not about the model. The gap to the exact value cannot be read as correlation, because the resolution error exceeds the quantity it would explain. And we decline to choose the σ that lands on Hartree–Fock: σ is fixed by hydrogen, and re-fixing it on helium would be the one thing this paper is organised to avoid.

A cross-check worth recording. At $\sigma = 0.2846$ the homogeneous treatment gives -3.022386 and the split treatment -3.026749—0.15% apart, from two different codes and two different constructions. That is a useful validation of both implementations, and it also shows that the distinction between them does *not* appear in helium's energy. Which is the point of the next paragraph: the difference is structural.

The split is worth keeping despite over-binding, and not as a variant to be averaged with the other. A spherical shared shape has no internal structure to orient, so it can generate no attraction between two such atoms at all; two territories can. Whether that mechanism accounts for the observed weak binding of He_2 we do not compute here, and we are careful not to claim it—dispersion-bound systems have been treated as outside this construction's range—but the *qualitative* difference is structural rather than numerical: one construction admits the mechanism and the other forbids it. That is a reason to retain both for helium, and a reason not to read the two energies as a bracket around a single answer.

The weak split is not yet measurable, and we report that rather than a number. Every attempt returns a state that fails the sphericity test: a computed dipole of 0.514 D where symmetry requires identically zero. By the standard set out in §7—a symmetric system's dipole is a pure error measurement needing no reference, and is the one check that cannot be argued with—such a state is a failed computation and not a result, so its energy is not quoted here.

The cause has been identified and it is an implementation defect rather than a property of the construction. At the kernel's own cell the relative coordinate vanishes, so the test that assigns cells to hemispheres ($\mathbf{r} \cdot \hat{\mathbf{a}} < 0$ against ≥ 0) excludes one sibling from that cell and grants it to the other. The softened attraction there is $Z/\varepsilon \approx 32\,E_h$ per unit density, so the two hemispheres are inequivalent *by construction*. Two controls support this and no other hypothesis: releasing the per-step enforcement reproduces the state *bit-identically*, so the asymmetry is fixed at initialisation; and rotating the dividing plane to a diagonal moves

the dipole by only 4.7%, so it is not an alignment of the plane with the grid. Neither control changes the kernel cell, which is why neither helps.

What helium does establish, independently of that defect. The one-electron ion on the same kernel at the same σ gives -2.053249 against the exact -2.0, 2.66% deep, with $T = 1.8855$ against 2.0. So the $Z = 2$ cusp and the transferred softening are *both* sound, and whatever limits helium is a two-electron matter. That is worth having: it rules out the calibration and the kernel as explanations before any two-electron number is in hand.

And the geometry is the reason to expect helium to be the harder case either way. Its interface passes through the kernel rather than between two of them, so it cuts the density maximum: $\int_\Gamma u^2 dS = 1.681$ against H_2's 0.496. Whatever error an interface treatment carries is weighted there by the largest density in the system.

6 HeH^+: the asymmetric case

Both systems above have interfaces fixed by symmetry, where the two constructions provably coincide, so neither can settle which formulation is right. The smallest two-electron system that can is HeH^+: two electrons on kernels of charge 2 and 1, net charge +1, equilibrium separation $R_e = 1.4632\, a_0$. Its interface lies at no symmetry plane, so amplitude matching and zero flux are genuinely different conditions there and the $\sim$5-fold discrepancy they show at a displaced interface should appear in an observable.

The construction is the natural one and needs nothing new: two domains, one electron each, one carrying a $Z = 2$ kernel and the other $Z = 1$. There is no pseudopotential, no core, no split ambiguity, and $r_c = 0$ throughout.

One confound must be declared, and it is not small. The two kernels have *different* charges, so a single softening σ cannot be calibrated for both: the value fixed on hydrogen is not the value a $Z = 2$ cusp requires, and the sharper cusp is the less well resolved. An error in HeH^+ is therefore attributable either to the interface asymmetry—the thing we want to measure—or to the Z mismatch, and the two are not separated by this system alone. The honest reading is that HeH^+ tests the formulation only to the extent that the σ transfer error is smaller than the effect sought, which has not been established.

The clean alternative, with equal kernels. The confound disappears if the asymmetry is put in the interface rather than in the nuclei: H_2 at fixed D with the dividing plane *displaced* off-centre, each domain still carrying unit charge. Both kernels are then $Z = 1$, one σ serves both, and the only asymmetry is the one under test. That measurement exists. Writing $\delta E = \frac{1}{2} k z_i^2$ for a displacement z_i,

h	k, continuity enforced	k, strict per-domain	ratio
0.250	0.7359	0.1296	5.68
0.125	0.7394	0.1472	5.02

The continuity stiffness is converged—0.5% across a halving—while the strict one is not, climbing 13.6% and still moving. That asymmetry is the sharper statement: the strict construction charges for a *discontinuity*, a mesh-scale object, so it has no well-defined interface stiffness in the continuum limit, whereas the continuity construction does. For an interface that must *move*—which is every molecule without a symmetry plane—that is a difference in kind rather than in accuracy.

So the programme is: HeH^+ for the observable, displaced H_2 for the mechanism. HeH^+ gives a real system with a real reference and a genuinely asymmetric interface, at the cost of the σ confound; displaced H_2 isolates the mechanism exactly but is not an observable configuration. Neither alone decides it; together they do, and the prediction to test is that the construction which converges in interface stiffness is the one whose geometry converges in HeH^+.

7 What this establishes, and what it does not

Not that the model is accurate at two electrons. It is not. On H_2 the two forms disagree by sixty-odd points of D_e, and on helium only one of them yields a state that passes the sphericity test at all. That width is the expected consequence of a declared property of the construction—the partition makes overlap impossible while the Coulomb term alone resists compression, and between equivalent domains across a mirror plane the partition contributes nothing—rather than a numerical failing to be driven down. The testable claim is that the error is *systematic*: one sign, one scale, and cancelling in differences taken at fixed composition.

And not that either observable has been predicted twice. One form places H_2's bond within 3.7% and the other its binding energy within 7.7%, from a single calibration performed on one electron. That is the paper's positive result, and its limit is stated in §4: the pair localises the deficiency to the interface treatment, and only the bond comes with a reason—H_2's Γ being fixed by symmetry—for preferring the form that gets it right.

A caveat on the references. The figure -1.174470 for H_2 is itself a *computed* quantity rather than a measured one, and the total electronic energy is in any case not an observable. The observables here are the dissociation energy, the bond length, and a dipole that must vanish by symmetry. We report total energies because they are what the two constructions differ in, not because they are measurable.

What the dipole buys. For a symmetric system the computed dipole must be identically zero, so any nonzero value is a pure error measurement needing no reference of any kind. It is the one check in this paper that cannot be argued with, and it is what disqualified the combined construction of §3 and would disqualify any future variant that improved the energy while losing the symmetry.